NETVERSE / ARCHITECTURE PLAN

PROPOSED · 26 OCT 2026

Three tiers. Two Availability Zones.

Repeat the web and application tiers in both zones. Give the database a managed failover partner.

REQUEST FLOW

Browser — HTTPS · TCP 443 → **Public ALB** — One service · two AZs → **1 · Web tier** — Frontend + API proxy → **Internal ALB** — One service · two AZs → **2 · App tier** — Business logic + API → **3 · Database** — RDS endpoint → primary

DNS resolves the public ALB name before the request. NAT handles separate outbound connections.

AWS REGION · us-east-1 — ONE VPC · 10.0.0.0/16 — Internet gateway · VPC attachment

Availability Zone A — us-east-1a

PUBLIC ENTRY 10.0.0.0/24

- Public ALB node A — Same public ALB across both zones
- NAT gateway A — EIP · outbound IPv4

1 · PRIVATE WEB 10.0.10.0/24

- Web server A — 10.0.10.10 — Frontend + API proxy · member of web-asg

2 · PRIVATE APPLICATION 10.0.20.0/24

- Internal ALB node A — Same internal ALB across both zones
- ↓
- Application server A — 10.0.20.10 — Business logic + API · member of app-asg

3 · ISOLATED DATABASE 10.0.30.0/24

- RDS PostgreSQL · primary — Serves reads and writes · initial placement

Availability Zone B — us-east-1b

PUBLIC ENTRY 10.0.1.0/24

- Public ALB node B — Same public ALB across both zones
- NAT gateway B — EIP · outbound IPv4

1 · PRIVATE WEB 10.0.11.0/24

- Web server B — 10.0.11.10 — Frontend + API proxy · member of web-asg

2 · PRIVATE APPLICATION 10.0.21.0/24

- Internal ALB node B — Same internal ALB across both zones
- ↓
- Application server B — 10.0.21.10 — Business logic + API · member of app-asg

3 · ISOLATED DATABASE 10.0.31.0/24

- RDS PostgreSQL · standby — Synchronous copy · promoted on failover

sync

ONE RDS DNS ENDPOINT · Both app replicas connect to the current primary · TCP 5432 + TLS

One web Auto Scaling group + one application Auto Scaling group span both AZs. Start with one instance per tier per AZ.

The database uses one Multi-AZ DB instance deployment. The standby does not serve application reads before promotion.

PUBLIC SUBNETS

0.0.0.0/0 → Internet gateway

One shared public route table

PRIVATE WEB + APP

0.0.0.0/0 → NAT in the same AZ

Separate private route table per AZ

ISOLATED DATABASE

10.0.0.0/16 → local only

No internet default route

IF AZ A FAILS — Healthy B web/app targets serve requests; RDS promotes B and updates the endpoint.

Allow for detection, database reconnection, and recovery time. Size surviving capacity for the required traffic.

Plan only · Example CIDRs and private IPs · ALBs may select healthy targets in either zone · Public access is limited to the public ALB

Based on AWS architecture, VPC, ALB, Auto Scaling, and RDS documentation. Sources and implementation steps: docs/AWS_THREE_TIER_PLAN.md